

YUANPENG LI

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EDUCATION

University of California, Irvine

Doctor of Philosophy (PhD) in Statistics

Irvine, CA

Sep. 2024 – Present

Jilin University – School of Mathematics

Bachelor of Science Degree, Major in Mathematics and Applied Mathematics

Changchun, China

Aug. 2020 – Jul. 2024

MANUSCRIPT

TOPPO: Rethinking PPO for Multi-Task Reinforcement Learning with Critic Balancing

Under review, NeurIPS 2026

Yuanpeng Li, Gefei Lin, Annie Qu, Rui Miao

RESEARCH EXPERIENCE

TOPPO: Rethinking PPO for Multi-Task Reinforcement Learning with Critic Balancing

Irvine, CA

First author; Advisors: Prof. Annie Qu, Dr. Rui Miao (corresponding)

Jul. 2023 – Present

- Diagnosed critic-side gradient ill-conditioning as a previously overlooked bottleneck of PPO in multi-task reinforcement learning, where tail tasks stall while easy tasks dominate value-function updates.
- Designed *Critic Balancing* for PPO — per-task PopArt value normalization, pre-activation LayerNorm in the critic body, and per-side gradient combiners (PCGrad / CAGrad / FairGrad, chosen independently for actor and critic) — to recondition gradients without enlarging the model.
- On Meta-World+ MT50, surpassed published SAC-family and ARS-family baselines on both mean and worst- k tail-task success while using up to $22.7\times$ fewer parameters and substantially fewer environment steps.

TEACHING

Graduate Teaching Assistant

Irvine, CA

UC Irvine, Department of Statistics

Sep. 2024 – Present

- STATS 7 — Basic Statistics (introductory inferential statistics: CIs, hypothesis testing, regression).
- STATS 67 — Introduction to Probability and Statistics for Computer Science.
- STATS 110 — Statistical Methods for Data Analysis I (ANOVA, simple and multiple linear regression).
- STATS 120C — Introduction to Probability and Statistics III (likelihood-ratio tests, ANOVA, regression inference and diagnostics).
- STATS 205P — Bayesian Data Analysis (Master of Data Science program; prior specification, Bayesian linear/GLM regression).

INTERNSHIP EXPERIENCE

Synkrotron

Xi'an, China

Intern, AI Research Department

Dec. 2022 – Jan. 2023

- Controlled remote Linux devices through FRP to configure network information in the automated road patrol project.
- Summarized the history of AI development in the artificial intelligence science teaching project, including algorithm development related to computer vision, history of test datasets, core tasks of computer vision, and common deep learning frameworks.
- Researched the use of computer-generated simulation datasets to enhance real-world performance in autonomous driving algorithm testing projects.

Hua Chuang Securities

Shenzhen, China

Equity Research Intern, Industry Research Division

Jul. 2022 – Oct. 2022

- Collected business information and data of six companies and built corresponding databases to support investment decisions.
- Utilized database data to estimate potential growth in the stock market.
- Updated institutional clients on Hua Chuang's research outputs instantly and constantly.
- Assisted team members in searching for papers and dealing with computer technical problems.

AWARDS

- Scholarship in the academic year 2022-2023 Oct. 2023
- The Chinese Mathematics Competitions: **National 3rd Prize** Dec. 2021
- Jilin Province Mathematics Competitions: **Provincial 2nd Prize** Dec. 2021

SKILLS

- **Programming Languages:** Python, C, MATLAB, Wolfram Mathematica, $\text{\LaTeX}$
- **ML Frameworks:** PyTorch, Tianshou, StableBaselines3, Gymnasium, MetaWorld, WandB, Scikit-learn, Pandas
- **AI Coding Tools:** Claude Code, OpenAI Codex CLI, GitHub Copilot, Cursor
- **Systems & DevOps:** Linux Server Administration, Git, Docker, VMware ESXi, OpenWrt, Network Administration